CBSE 2027 Physics — Predicted Paper (Set 8 of 10)

SECTION A: Objective Type Questions (16 Marks)

This section consists of 12 MCQs and 4 Assertion-Reasoning questions of 1 mark each.

Q1. A soap bubble is given a negative charge. Its radius will: (a) decrease (b) increase (c) remain unchanged (d) fluctuate *(Chapter: Electric Charges and Fields)* | **[Confidence: Very High]**

Q2. A parallel plate capacitor is charged by a battery and then the battery is disconnected. If the plates are slowly pulled apart, the potential difference across the plates will: (a) increase (b) decrease (c) remain the same (d) become zero *(Chapter: Electrostatic Potential and Capacitance)* | **[Confidence: High]**

Q3. Two wires A and B of the same material and equal length have their radii in the ratio $1:2$. The ratio of their electrical resistances ($R_A : R_B$) is: (a) $1:2$ (b) $2:1$ (c) $1:4$ (d) $4:1$ *(Chapter: Current Electricity)* | **[Confidence: Very High]**

Q4. A proton enters a uniform magnetic field with a velocity v at an angle of 60° to the direction of the field. The path followed by the proton is a helix. The pitch of the helix is: (a) $\frac{2\pi mv}{qB}$ (b) $\frac{\pi mv}{qB}$ (c) $\frac{\sqrt{3}\pi mv}{qB}$ (d) $\frac{2\pi mv \sin 60^\circ}{qB}$ *(Chapter: Moving Charges and Magnetism)* | **[Confidence: High]**

Q5. Which of the following is an example of a paramagnetic substance? (a) Copper (b) Bismuth (c) Aluminum (d) Water *(Chapter: Magnetism and Matter)* | **[Confidence: Very High]**

Q6. Two coils have a mutual inductance of $0.005\ \mathrm{H}$. The alternating current changes in the first coil according to the equation $I = 10\sin(100\pi t)$. The maximum value of the emf induced in the second coil is: (a) $2\pi\ \mathrm{V}$ (b) $5\pi\ \mathrm{V}$ (c) $10\pi\ \mathrm{V}$ (d) $50\pi\ \mathrm{V}$ *(Chapter: Electromagnetic Induction)* | **[Confidence: High]**

Q7. In a step-up transformer, which of the following physical quantities strictly decreases from the primary to the secondary coil? (a) Voltage (b) Current (c) Power (d) Frequency *(Chapter: Alternating Current)* | **[Confidence: Very High]**

Q8. Which of the following electromagnetic waves has the lowest frequency? (a) Microwaves (b) Infrared (c) Radio waves (d) X-rays *(Chapter: Electromagnetic Waves)* | **[Confidence: Very High]**

Q9. An air bubble inside a glass slab ($\mu = 1.5$) appears to be $6\ \mathrm{cm}$ deep when viewed from one face and $4\ \mathrm{cm}$ deep when viewed from the opposite face. The actual thickness of the glass slab is: (a) $10\ \mathrm{cm}$ (b) $6.67\ \mathrm{cm}$ (c) $15\ \mathrm{cm}$ (d) $10.5\ \mathrm{cm}$ *(Chapter: Ray Optics and Optical Instruments)* | **[Confidence: Medium]**

Q10. In Young's Double Slit Experiment, the fringe width in air is β. If the entire experimental arrangement is immersed in a transparent liquid of refractive index μ, the new fringe width becomes: (a) $\mu\beta$ (b) β/μ (c) $\beta/(\mu-1)$ (d) β *(Chapter: Wave Optics)* | **[Confidence: Very High]**

Q11. The de-Broglie wavelength (λ) of a non-relativistic particle of mass m and kinetic energy E is given by: (a) $h/\sqrt{2mE}$ (b) $\sqrt{2mE}/h$ (c) $h/2mE$ (d) $hE/2m$ *(Chapter: Dual Nature of Radiation & Matter)* | **[Confidence: Very High]**

Q12. In the nuclear reaction $^{14}_{7}\mathrm{N} + ^{4}_{2}\mathrm{He} \rightarrow ^{17}_{8}\mathrm{O} + X$, the unknown particle X is a/an: (a) Electron (b) Proton (c) Neutron (d) Positron *(Chapter: Nuclei)* | **[Confidence: High]**

Q13. Assertion (A): The capacitance of a parallel plate capacitor increases when a thick conducting slab is introduced between its plates. **Reason (R):** The electric field inside the conducting slab is zero, which effectively decreases the potential difference between the plates for a given charge. *(Chapter: Electrostatic Potential and Capacitance)* | **[Confidence: Very High]**

Q14. Assertion (A): The drift velocity of free electrons in a metallic conductor decreases with an increase in temperature. **Reason (R):** With an increase in temperature, the collision frequency of electrons with the lattice ions increases, which decreases the average relaxation time. *(Chapter: Current Electricity)* | **[Confidence: Very High]**

Q15. Assertion (A): Lenz's law violates the principle of conservation of energy. **Reason (R):** Induced emf always opposes the change in magnetic flux responsible for its production. *(Chapter: Electromagnetic Induction)* | **[Confidence: High]**

Q16. Assertion (A): In a p-n junction diode, the reverse saturation current is very small and almost independent of the applied reverse bias voltage (up to the breakdown voltage). **Reason (R):** The reverse current is strictly due to the drift of minority charge carriers across the junction, which is limited by the rate of thermal generation, not the applied voltage. *(Chapter: Semiconductor Electronics)* | **[Confidence: Very High]**

SECTION B: Very Short Answer Type Questions (10 Marks)

This section consists of 5 questions of 2 marks each.

Q17. Define electric field intensity at a point. Two point charges $+q$ and $-q$ are placed at a distance $2a$ apart in a vacuum. Calculate the magnitude and direction of the net electric field at the exact midpoint of the line joining them. *(Chapter: Electric Charges and Fields)* | **[Confidence: Very High]**

Q18. A circular coil of N turns and radius R carries a steady current I. It is unwound and rewound to make a new coil of radius $R/2$. If the same current I is passed through it, find the ratio of the magnetic field at the center of the new coil to that of the original coil. *(Chapter: Moving Charges and Magnetism)* | **[Confidence: High]**

Q19. A plane electromagnetic wave travels in a vacuum along the positive z-direction. What can you say about the directions of its oscillating electric and magnetic field vectors? If the frequency of the wave is $30\ \mathrm{MHz}$, what is its wavelength? *(Chapter: Electromagnetic Waves)* | **[Confidence: Very High]**

Q20. Draw a graph showing the variation of stopping potential (V_0) with the frequency (ν) of the incident radiation for two different photosensitive materials having work functions W_1 and W_2 ($W_1 > W_2$). What does the ratio of the slopes of these two graphs equal? *(Chapter: Dual Nature of Radiation and Matter)* | **[Confidence: High]**

Q21. Calculate the ratio of the maximum wavelength of the Lyman series to the maximum wavelength of the Balmer series in the hydrogen emission spectrum. *(Chapter: Atoms)* | **[Confidence: Medium]**

SECTION C: Short Answer Type Questions (21 Marks)

This section consists of 7 questions of 3 marks each.

Q22. State Gauss's Law in electrostatics. A point charge $+q$ is placed exactly at the center of a hollow cube of side L. (a) What is the total electric flux passing through the entire cube? (b) What is the electric flux passing through any one face of the cube? (c) If the charge $+q$ is moved to one of the corners of the cube, what will be the new total flux passing through the cube? *(Chapter: Electric Charges and Fields)* | **[Confidence: Very High]**

Q23. Define the temperature coefficient of resistivity. A heating element using nichrome connected to a 230 V supply draws an initial current of 3.2 A which settles after a few seconds to a steady value of 2.8 A. What is the steady temperature of the heating element if the room temperature is 27°C? (Temperature coefficient of resistance of nichrome averaged over the temperature range is $1.7 \times 10^{-4}\ ^\circ\text{C}^{-1}$). *(Chapter: Current Electricity)* | **[Confidence: Very High]**

Q24. Two identical circular coils, each of radius R and carrying the same current I, are kept in parallel planes having a common axis passing through their centers. The direction of current in both coils is the same. Find the magnitude of the net magnetic field at the midpoint of the axis between them if they are separated by a distance R. *(Chapter: Moving Charges and Magnetism)* | **[Confidence: High]**

Q25. Define the term mutual inductance. Two concentric circular coils of radii R and r ($R \gg r$) are placed coaxially with their centers coinciding. Derive a mathematical expression for the mutual inductance (M) of this combination. *(Chapter: Electromagnetic Induction)* | **[Confidence: Very High]**

Q26. State the two essential conditions for total internal reflection to occur. Derive the relation between the critical angle and the refractive index of the medium. A ray of light is incident normally on one face of a right-angled isosceles glass prism. It undergoes total internal reflection at the hypotenuse. Find the minimum required refractive index of the glass. *(Chapter: Ray Optics and Optical Instruments)* | **[Confidence: Very High]**

Q27. Consider two coherent light waves represented by $y_1 = a\sin(\omega t)$ and $y_2 = a\sin(\omega t + \phi)$ superimposing at a point. Derive the expression for the resultant intensity at that point. Hence, state the conditions (in terms of phase difference ϕ) for constructive and destructive interference. *(Chapter: Wave Optics)* | **[Confidence: Very High]**

Q28. Define the terms 'mass defect' and 'binding energy' of a nucleus. Draw a plot showing the variation of binding energy per nucleon (BE/A) versus the mass number (A) for a large number of nuclei. Explain the release of energy in the phenomena of nuclear fission and fusion using this plot. *(Chapter: Nuclei)* | **[Confidence: Very High]**

SECTION D: Case-Study Based Questions (8 Marks)

This section consists of 2 questions of 4 marks each. Questions usually have internal sub-parts.

Q29. Case Study 1: LCR Circuit and Sharpness of Resonance *Passage:* In a series LCR circuit connected to an AC source of variable frequency, the current amplitude is maximum at a specific frequency called the resonant frequency (ω_r). At this state, the inductive reactance perfectly cancels the capacitive reactance, making the circuit purely resistive. The sharpness of this resonance is measured by the Quality factor (Q-factor). A higher Q-factor indicates a sharper, narrower resonance peak, which is highly desirable in tuning circuits of radios and televisions. *(a)* What is the phase difference between the applied voltage and the current at resonance? (1 mark) *(b)* Write the formula for the resonant angular frequency (ω_r) of the circuit. (1 mark) *(c)* A series LCR circuit with $L = 2.0\ \mathrm{H}$, $C = 32\ \mu\mathrm{F}$, and $R = 10\ \Omega$ is connected to a variable frequency $200\ \mathrm{V}$ AC supply. Calculate the resonant angular frequency and the Q-factor of the circuit. **OR** Prove mathematically that at the condition of resonance, the net voltage drop across the inductor-capacitor (LC) series combination is always zero. (2 marks) *(Chapter: Alternating Current)* | **[Confidence: High]**

Q30. Case Study 2: Rectifiers and Diodes *Passage:* A rectifier is an essential electronic device that converts alternating current (AC) into direct current (DC). It utilizes the core property of a p-n junction diode, which allows electric current to pass only when it is forward-biased and strongly blocks it when reverse-biased. A half-wave rectifier uses a single diode to rectify only one half of the AC cycle, while a full-wave rectifier uses two center-tapped diodes (or a bridge) to rectify both halves of the cycle, yielding a smoother output. *(a)* If the frequency of the input AC voltage is $50\ \mathrm{Hz}$, what is the frequency of the output voltage of a half-wave rectifier? (1 mark) *(b)* Why is a large filter capacitor often connected in parallel to the load resistance in the output circuit of a rectifier? (1 mark) *(c)* Draw a neat, labelled circuit diagram of a full-wave rectifier using a center-tapped transformer. **OR** Draw the input and output voltage waveforms for a full-wave rectifier. If the input frequency is $50\ \mathrm{Hz}$, what is the fundamental ripple frequency of its output? (2 marks) *(Chapter: Semiconductor Electronics)* | **[Confidence: Very High]**

SECTION E: Long Answer Type Questions (15 Marks)

This section consists of 3 questions of 5 marks each. All questions have an internal "OR" choice.

Q31. (a) Derive an expression for the total electrostatic potential energy of a system of two point charges q_1 and q_2 located at position vectors $\vec{r_1}$ and $\vec{r_2}$ respectively, in the presence of an external electric field $\vec{E}$. **(b)** Three point charges $+q$, $+q$, and $-2q$ are placed at the three vertices of an equilateral triangle of side L in a region with no external electric field. Calculate the total work done required to completely separate the charges to infinity. **OR (a)** Define electric dipole moment. State its SI unit and direction. **(b)** Derive an expression for the electric field intensity at a point P located at a distance r on the axial line of an electric dipole of length $2a$. **(c)** An electric dipole of length 2 cm is placed at an angle of 30° with a uniform electric field of $2 \times 10^5 \text{ N/C}$. It experiences a torque of $8\sqrt{3} \text{ Nm}$. Calculate the magnitude of charge on the dipole and its potential energy in this orientation. *(Chapter: Electrostatics)* | **[Confidence: Very High]**

Q32. (a) Define the internal resistance of a cell. On what physical factors of the cell does it depend? **(b)** Two non-ideal cells of emfs E_1 and E_2 and internal resistances r_1 and r_2 respectively, are connected in parallel (positive terminal to positive terminal). Deduce the expressions for the equivalent emf (E_{eq}) and the equivalent internal resistance (r_{eq}) of the combination. **(c)** If $E_1 = 1.5 \text{ V}$, $r_1 = 0.2 \ \Omega$, $E_2 = 2.0 \text{ V}$, and $r_2 = 0.3 \ \Omega$, calculate the equivalent emf of this parallel combination. **OR (a)** Define the coefficient of self-inductance of a coil. Write its SI unit. **(b)** Derive an expression for the magnetic potential energy stored in an inductor of self-inductance L carrying a steady current I. **(c)** Two coils have a mutual inductance of 1.5 H. If the current in the primary circuit drops uniformly from 20 A to zero in 0.05 seconds, calculate the average magnitude of the emf induced in the secondary coil. *(Chapter: Current Electricity / Electromagnetic Induction)* | **[Confidence: Very High]**

Q33. (a) Draw a clear, labelled ray diagram to show the formation of an image by a concave mirror when an object is placed between the center of curvature (C) and the principal focus (F). **(b)** Using the above ray diagram, derive the mirror formula: $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$. State the new Cartesian sign conventions used. **(c)** A mobile phone lies flat along the principal axis of a concave mirror such that its screen faces the mirror. Explain qualitatively why the magnification of the phone is not uniform across its entire length. **OR (a)** Describe the diffraction of light due to a single narrow slit of width 'a'. Explain the formation of the central maximum and the secondary minima on the screen using Huygens' principle of secondary wavelets. **(b)** Write the mathematical expression for the angular width of the central maximum. **(c)** How does the angular width of the central maximum change if: (i) the slit width 'a' is decreased, and (ii) the monochromatic light is replaced by light of a shorter wavelength? Justify. *(Chapter: Ray Optics / Wave Optics)* | **[Confidence: Very High]**
